

Komeil Najafi

Curriculum Vitae

Isfahan, Iran
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Education

- 2022–2025 **M.Sc. in Physics**, *Isfahan University of Technology*, Isfahan, Iran.
2018–2022 **B.Sc. in Physics**, *Isfahan University of Technology*, Isfahan, Iran.

Research Interests

Quantum Technology Quantum information, Quantum sensing, Optomechanics, spin systems, Quantum computing, Quantum Photonics, Silicon Nitride Waveguides, Lithium Niobate Waveguides

Publications

- In Progress **Sensing Gravitational Gradients with a Magnonic System.**
We're using magnons, the quanta of spin waves, in a YIG sphere placed in a microwave cavity to detect gravity gradients by sensing gravitational differences, which can be read from the microwave cavity.

Projects

- 2022–2025 **M.Sc. Project**, *Supervisor: Dr. M. Zarei.*
Sensing Gravitational Gradients with a Magnonic System.
2018–2022 **B.Sc. Project**, *Supervisor: Dr. F. Loran.*
Studies on Friedman's space-time.

Research Assistant

- Sep 2025 – Present **Research Assistant at the Center for Quantum Science and Technology (CQST)**, *Isfahan University.*
Working on the fabrication and characterization of integrated photonic devices based on lithium niobate (LiNbO_3). Hands-on cleanroom experience in:
 - Lithography techniques: direct-write lithography and UV-mask lithography for optical waveguide fabrication.
 - Poling setups: periodic poling of $\text{MgO}:\text{LiNbO}_3$ using both metallic-electrode and liquid-electrolyte methods, targeting high-quality domain inversion for SPDC and entangled photon generation.
 - Dissipation setups: free-space measurements of propagation loss in fabricated waveguides at 1550 nm, including thermal stability characterization with hot-plate heating.
 - Dissipation setups: free-space measurements of propagation loss in fabricated waveguides at 1550 nm, including thermal stability characterization with hot-plate heating.
 - Integrated photonic SIN design setups: design and optimization of power dividers and directional couplers in dielectric waveguides, employing neural-network-assisted inverse design to improve coupling efficiency, bandwidth, and fabrication tolerance at 1550 nm.

Work Experience

- Fall 2020 **Android Application Developer.**
Developed an educational game app for children.
2019–2020 **Independent Property Manager.**
Managed villa rentals similar to Airbnb.

Conferences and Workshops Organized

- Fall 2024 **Winter School of Quantum Technologies**, *Organizing Committee Member, IUT.*
Spring 2023 **National Conference on Quantum Technologies**, *Organizing Committee Member, IUT.*

Conferences and Workshops Attended

- Summer 2024 **Summer School on Quantum Chaos, Dissipation & Information**, *Nordita, Stockholm.*
Poster: Gravity Gradient Sensing via Magnonic Systems.
Summer 2024 **International Conference of Physics Students (ICPS)**, *Georgia.*
Talk: Quantum Gravity Gradiometers.
Winter 2024 **Second Workshop on Quantum Technology**, *Sharif University of Technology, Tehran.*
Winter 2024 **Tensor Networks for Many-body Systems**, *IUT, Isfahan.*
Winter 2024 **The First Winter School of Quantum Technologies**, *IUT, Isfahan.*
Spring 2023 **Introductory Quantum Computing (with QuTiP)**, *Sharif University of Technology, Isfahan.*

Online Courses

- Winter and Spring 2024 **Comprehensive Quantum Computing**, *Aria Quanta.*
Quantum simulation, Qiskit, QAOA, quantum ML, superconducting/photonic/ion qubits.
Winter 2024 **Exploring Quantum Physics**, *University of Maryland (Coursera).*
Summer 2023 **Quantum Optics 1 & 2**, *École Polytechnique (Coursera).*
Winter 2021 **Advanced Android Programming**, *Faradars.org.*

Internship

- July 2025 – **Internship at the Center for Quantum Science and Technology (CQST)**, *Isfahan University.*
Sep 2025 Ongoing internship at CQST, working on lithography-based techniques including direct-write lithography, periodically poled lithium niobate (PPLN), and electric field poling for photonic device fabrication.

Computer Skills

- Programming Python, Java, C++, XML, HTML/CSS, Android, Kotlin ,neural-network
Software Word, PowerPoint, Excel, LaTeX, Mathematica, Comsol, Zemax

Teaching Experience

- Fall 2023 **Teaching Assistant**, *Electricity Laboratory, Isfahan University of Technology.*

Honors and Awards

- Spring 2024 **Mobarakeh Steel Company Scholarship.**
Awarded for significant research contributions with applications in the steel industry.

Languages

- Persian **Native**
English **Fluent**

TOEFL iBT: 89 (2025)

References

- Dr. Moslem Zarei Assistant Professor, IUT
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Dr. Farhad Shahbazi Associate Professor, IUT
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Dr. Keyvan Associate Professor, IUT
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